EXP-6 : To infer an affine key from the frequent ciphertext letters B and U.

PROGRAM :

Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
int main(void) {
for (int a = 1; a < 26; a += 2) {
if (a == 13) continue;
for (int b = 0; b < 26; b++)
if ((4*a+b)%26 == 1 && (19*a+b)%26 == 20) {
int inv = 1;
while ((a*inv)%26 != 1) inv++;
printf("a = %d, b = %d\n", a, b);
printf("P = %d(C - %d) mod 26\n", inv,b);
char c[] = "BU";
printf("Sample BU -> ");
for (int i = 0; c[i]; i++)
putchar((inv*(c[i]-'A'-b+26))%26+'A');
puts("");
}
}
return 0;
}
```

OUTPUT :

```
a = 3, b = 15
P = 9(C - 15) mod 26
Sample BU -> ET


...Program finished with exit code 0
Press ENTER to exit console.
```